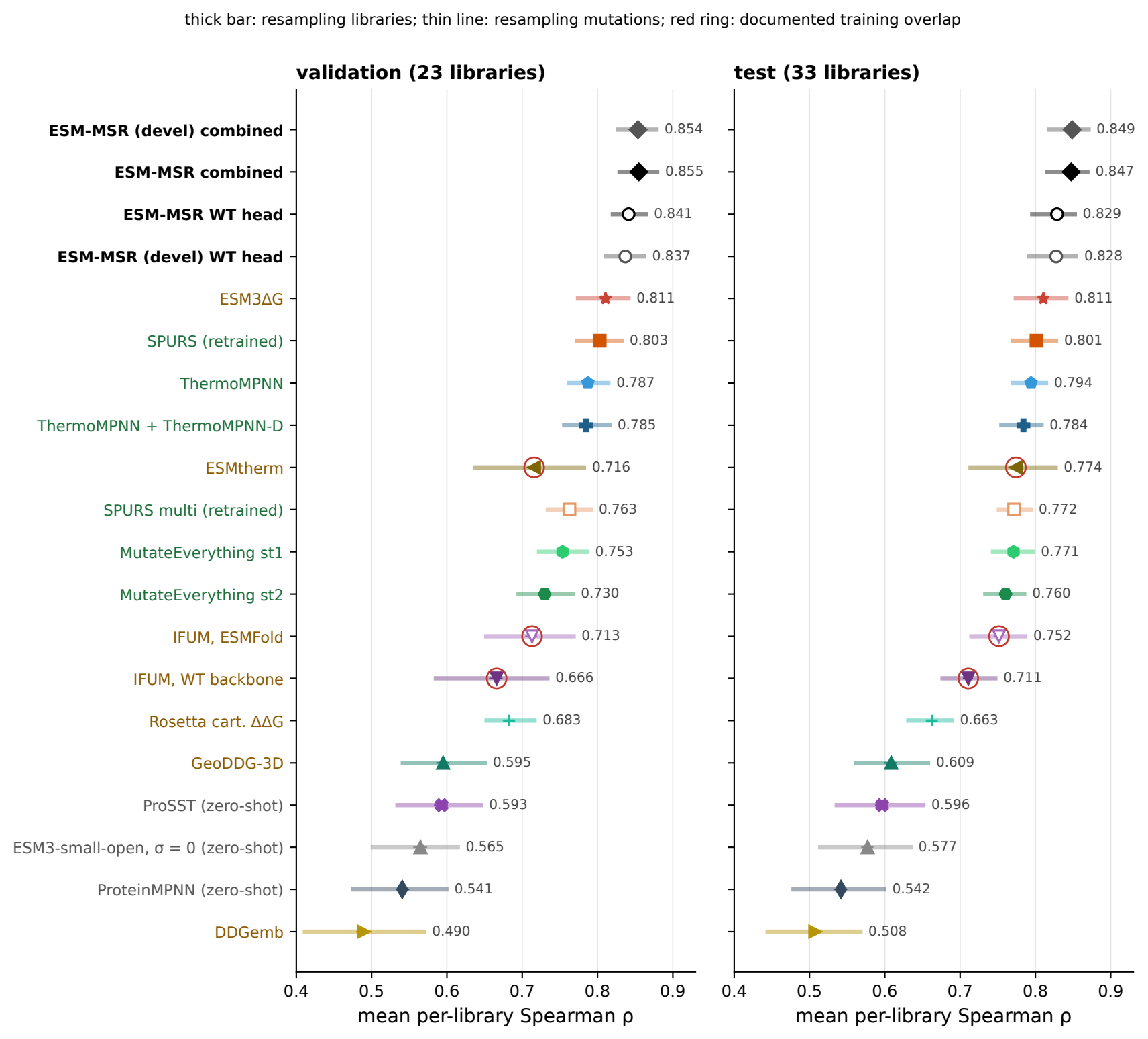
thick bar: resampling libraries; thin line: resampling mutations; red ring: documented training overlap
validation (23 libraries)
test (33 libraries)
ESM-MSR (devel) combined
ESM-MSR combined
ESM-MSR WT head
ESM-MSR (devel) WT head
ESM3ΔG
SPURS (retrained)
ThermoMPNN
ThermoMPNN + ThermoMPNN-D
ESMtherm
SPURS multi (retrained)
MutateEverything st1
MutateEverything st2
IFUM, ESMFold
IFUM, WT backbone
Rosetta cart. ΔΔG
GeoDDG-3D
ProSST (zero-shot)
ESM3-small-open, σ = 0 (zero-shot)
ProteinMPNN (zero-shot)
DDGemb
0.854
0.855
0.841
0.837
0.811
0.803
0.787
0.785
0.716
0.763
0.753
0.730
0.713
0.666
0.683
0.595
0.593
0.565
0.541
0.490
0.849
0.847
0.829
0.828
0.811
0.801
0.794
0.784
0.774
0.772
0.771
0.760
0.752
0.711
0.663
0.609
0.596
0.577
0.542
0.508
0.4
0.5
0.6
0.7
0.8
0.9
mean per-library Spearman ρ